

Smart Bidding Based Agri Trade Platform for the Fairness of Farmers

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Abstract—Smart Bidding Based Agri-Trade Platform is build to solve the problem of unfair prices and not gaining access to the market by the smallholder farmers by offering farmers a digital platform of offering their products at a fixed price or an auction. This enables the farmers to be more in control of their sales strategy and have direct access to buyers hence less reliance in intermediaries. Product listing tools are simple and easy to use the platform has smart inventory management and real-time market research, thus the farmer is able to achieve maximum sale and minimize waste. It promotes sustainability in terms of the higher economic self-sufficiency and resilience of communities. The platform creates farm profits that are stable because of the ability of the farmer to maintain larger value of their produce. Nevertheless, the platform is faced with issues such as the digital divide that needs specific training programs and infrastructure improvement. Conclusively, The Platform will be a key milestone in the modernization of agricultural trade, its sustainability in environmental, social and economic contexts together with the technical side of it and livelihoods of the smallholder farmers. Future work will be on streamlining the features of the platform and dispelling the challenges of adoption in order to get the most out of it.

Index Terms—Agricultural trade platform Smart bidding, Smallholder farmers, Fair pricing, Sustainability, Digital agriculture, E-commerce

I. INTRODUCTION

The agricultural sector in Bangladesh faces challenges like unfair pricing, limited market access, and waste due to inefficient supply chains. The Smart Bidding Based Agri Trade Platform connects smallholder farmers directly to buyers through flexible selling options and smart inventory management. This platform promotes fair pricing, market access, and sustainability across environmental, social, and economic dimensions.

This study introduces the platform as a solution to reduce intermediary exploitation, maximize profits for farmers, and provide access to broader markets. Key contributions include offering flexible selling options [1], [2], enhancing market access with mobile compatibility [3], [4], and promoting

sustainability through smart inventory management and middlemen integration [5], [6].

The research addresses challenges like limited digital inclusion and the lack of localized solutions in Bangladesh's agricultural context. It aims to fill gaps in sustainability and middlemen roles while offering a tailored, inclusive solution for the country's farmers.

The study addresses the following research questions:

- How can a digital platform integrate various selling mechanisms to maximize profits for smallholder farmers while ensuring sustainability?
- What strategies can overcome the digital divide to ensure inclusive access for all farmers in Bangladesh?
- How can smart inventory management and real-time market analysis enhance profitability and reduce waste in the supply chain?

The objectives are to:

- Design a user-friendly platform with diverse selling options tailored to smallholder farmers.
- Implement features promoting sustainability, including waste reduction and market efficiency.
- Evaluate the platform's impact on farmer incomes, market access, and sustainability.

The Smart Bidding Based Agri Trade Platform uses open-source technologies to offer flexible selling options (fixed-price, auctions, and advance sales) for farmers to maximize profits. It optimizes inventory using agent-based modeling, reduces waste, and provides real-time pricing insights for better decision-making [6]. The platform integrates middlemen for logistics, ensuring fair pricing and addressing social and economic sustainability. Mobile compatibility, Bangla content, and secure payment systems ensure accessibility, trust, and quality, aiming to transform agricultural trading in Bangladesh and empower smallholder farmers.

The platform promotes environmental sustainability by minimizing waste through smart inventory management. Social sustainability is achieved by providing farmers access to broader markets and fair pricing. Economic sustainability is supported by flexible selling options, boosting profitability. Technical sustainability is ensured with scalable, open-source technologies and mobile compatibility for long-term accessibility.

LITERATURE REVIEW

The Smart Bidding Agri-Trade Platform connects farmers to markets through fixed-price sales, auctions, and smart inventory management, addressing challenges like unfair pricing and limited access. This review synthesizes insights from 19 papers, exploring digital platforms, auction mechanisms, and e-commerce in agriculture to enhance pricing, market access, and sustainability.

Related Works

The reviewed studies collectively underscore the transformative potential of digital platforms and auction mechanisms in agricultural trading. Das [1] models collusion in Indian agricultural auctions, emphasizing transparent bidding to prevent price manipulation. Levi et al. [2] demonstrate a two-stage auction on India's Unified Market Platform, increasing farmer prices by 3.6%, benefiting over 10,000 farmers. Pesci et al. [3] point to an online sale disparity in California farmers during the COVID19 pandemic mentioning such obstacles as the lack of internet penetration. Ma et al. [4] emphasize e-commerce's role in overcoming market access barriers, enhancing farmer incomes. Miyashita [5] presents an online double auction of perishable goods where the failure of trade was minimized with the time-sensitive bidding. Huang and Song [6] develop an agent-based model for inventory control, minimizing waste in agriculture supply chains. Mandhare et al. [7] design an online auction system for farmers, enabling fair price discovery. Banu [8] introduces AgriTrust, a blockchain-based bidding system for transparent trade. Fang et al. [9] study cloud farm platforms, optimizing farmer benefits for market linkages. To improve project success, Sadia et al. [10] present a process model for incremental asset building in software project management. Kumarathunga et al. [11] explore blockchain-based trust mechanisms, facilitating direct trade. Feng [12] examines small farmer e-tailers' exploitation in China, highlighting platform control issues. Ealias et al. [13] propose a dynamic bidding platform for e-commerce products. Alves Junior et al. [14] introduce a DEA-based method for bidding priority, optimizing pricing decisions. Bacarin et al. [15] present e-negotiation frameworks for sustainable contracts. Leduc et al. [16] propose a blockchain-based marketplace, enhancing transaction transparency. Vavekanand et al. [17] investigate e-commerce's potential for rural market expansion. Xie et al. [18] explore digital agriculture through cooperatives, enhancing market access. Sadia et al. [19] examine how a unified market platform enhances agricultural prices and farmer profitability by increasing competition and transparency.

Summary of Findings

The reviewed studies suggest digital platforms and auction mechanisms can enhance fair pricing, market access, and sustainability for smallholder farmers. They emphasize transparent bidding, innovative auctions, and direct buyer connections. However, challenges like digital divides and middlemen displacement persist, requiring inclusive solutions.

Gap Analysis

Despite advancements, gaps remain:

- **Digital Inclusion:** Limited internet access and digital literacy hinder adoption [3], [4], [11]. Mobile-friendly interfaces and training are needed.
- **Sustainability Integration:** Few platforms address environmental, social, and economic sustainability comprehensively [6]. Holistic approaches are required.
- **Middlemen Transition:** Lack of strategies for integrating middlemen [12]. The platform's delivery partner model addresses this.
- **Localized Solutions:** Platforms lack tailoring for Bangladesh [7], [18]. Localized interfaces are essential.

The platform fills these gaps with a localized, sustainable design.

II. RESEARCH METHODOLOGY

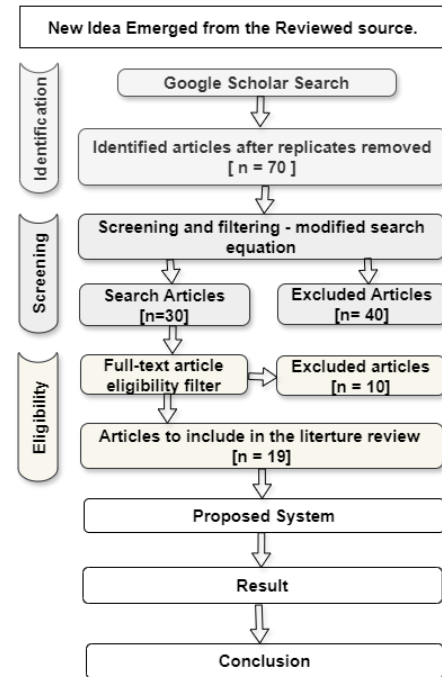


Fig. 1. Prisma Diagram and Research Methodology

Fig. 1. shows the process of selecting 19 articles from 70 for a literature review, leading to the proposed system, results, and conclusion.

This study assesses the effectiveness of the Smart Bidding Based Agri Trade Platform in addressing unfair pricing, market

access, and inventory management challenges faced by smallholder farmers in Bangladesh. The platform integrates selling mechanisms like fixed-price sales, auctions, and advance sales to maximize profits while ensuring sustainability. The study explores the use of Agile and hybrid methodologies for the platform's development, facilitating iterative progress and flexibility in addressing evolving needs and features. It also examines strategies to overcome the digital divide, ensuring inclusive access for all farmers, and highlights both the potential benefits of digital platforms and barriers, such as exploitation by intermediaries and limited technology adoption.

Method - Research Framework

The research validated various approaches, combining both quantitative and qualitative techniques. Quantitative data was collected through structured surveys administered to smallholder farmers using the platform, assessing their sales performance, income, and market access. Semi-structured interviews were conducted with farmers, platform administrators, and industry experts to gather qualitative insights into the platform's usability, challenges, and broader impacts. The quantitative data was evaluated using simple statistics and regression analysis, while the qualitative data was processed systematically to identify major themes and difficulties.

Sample Profile

The study focused on smallholder farmers who had access to the Smart Bidding Based Agri Trade Platform and were actively participating in the platform's sales processes. The sample included farmers from various agricultural sectors such as vegetables, fruits, and grains, selected from regions in Bangladesh where the platform had been implemented. A targeted sampling method was used to ensure a representative mix of farmers based on factors such as farm size, technological literacy, and geographical location.

Data Collection Methods

The sample consisted of smallholder farmers actively using the Smart Bidding Based Agri Trade Platform. Farmers were selected based on farm type, size, geographical location, and level of technological access to ensure a representative sample. Data collection methods included surveys for quantitative data on market performance and semi-structured interviews for qualitative insights.

Data Analysis Method

The quantitative data analysis included descriptive statistics to summarize key metrics and regression analysis to identify any correlations between platform use and income improvements. Qualitative data was analyzed using thematic analysis to extract patterns and insights regarding farmers' experiences and challenges.

Limitation of Research Design

However, several limitations were acknowledged in this research design. There was potential sampling bias due to the limited adoption of the platform in certain regions, and self-reported data could have introduced biases in the responses. Additionally, the study's cross-sectional nature limited its ability to capture the long-term impacts of the platform. The findings may also have limited generalizability to areas with limited internet access or technological infrastructure, where the platform's adoption may differ.

III. PROPOSED SYSTEM

The prototype agricultural platform, developed using HTML, CSS, Bootstrap, Node.js, and MongoDB, connects farmers directly with buyers to eliminate intermediaries and increase revenues. Node.js manages backend operations, including user authentication, order processing, and real-time bidding, while MongoDB stores user, product, and transaction data. Cloud infrastructure enables scalability, and SSL encryption ensures secure transactions. The platform's architecture is outlined through Rich Picture and High-Level Design, with Node.js and MongoDB guaranteeing efficient performance and secure data management.

A. Rich Picture

The proposed platform enables farmers to update their crop inventory in real time, allowing buyers to view and purchase products while automatically updating stock levels for seamless farm-to-buyer transactions. Crops online, buyers shop or bid, and products are stored and delivered via a warehouse and transport system.

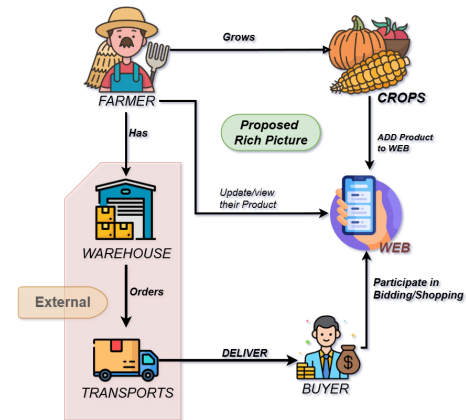


Fig. 2. Rich Picture

Fig.2. depicts a system where farmers list

B. High-Level Design

The High-Level Design of the agricultural platform integrates key subsystems for smooth operations. Farmers update their inventory via the Farmer Subsystem, which syncs with the Warehouse Subsystem. Buyers place orders through the

Buyer Subsystem, updating Inventory Management. The Warehouse processes orders, and Logistics handles delivery. The System Admin ensures overall platform management.

To implement this, we've developed key diagrams: Data Flow Diagrams (DFD) at both Context and Subsystems levels, Use Case Diagrams for user interactions, and Class, Sequence, and Activity Diagrams to define system structure and process flows. The Deployment Diagram illustrates the system's physical setup, ensuring efficiency, transparency, and better market access for farmers.

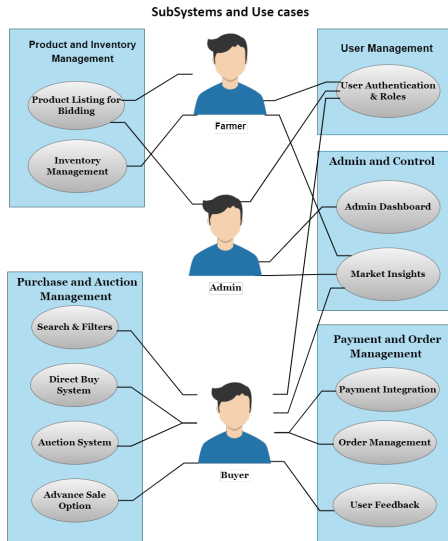


Fig. 3. Subsystem and Use Cases

Fig. 3 shows the use case and sub systems of the platform.

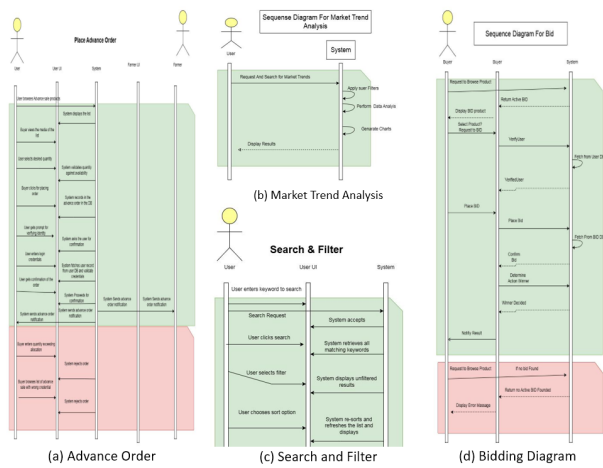


Fig. 4. Sequence Diagrams: a) Advance Order b) Market Trend Analysis c) Search and Filter d) Bidding Diagram

C. Implementation

The Implementation of the agricultural platform involves subsystems with specific inputs and outputs. Farmers update inventory in real-time, which is reflected across the platform.

Buyers place orders, triggering updates to stock levels, while the warehouse manages storage and fulfillment. Logistics handle deliveries, and the system admin oversees user management and platform settings.

In terms of Deployment, the system consists of web servers hosting the user interfaces for farmers, buyers, and admins, and a cloud-based database for storing all data. The Warehouse Management System operates locally, and the Logistics Server tracks deliveries. The Admin Interface is secured for platform management. The infrastructure is designed for scalability, with APIs ensuring smooth data flow and secure operations across subsystems.

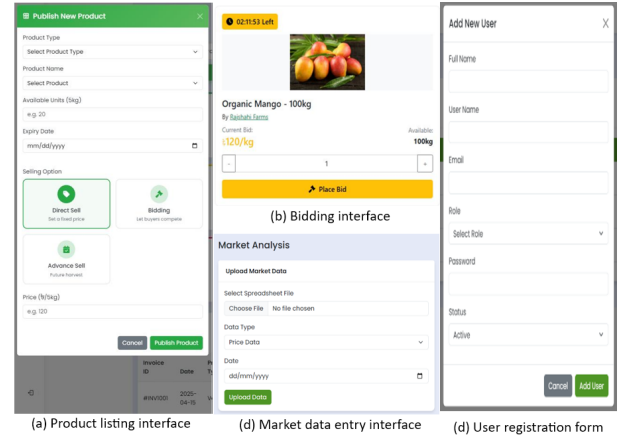


Fig. 5. Input Design Screen

Fig. 5. Feature-Specific Interface Screens: (a) Product listing interface for farmers showing various product types like bidding-based, direct sale, and advance sale; (b) Bidding interface for buyers to view and place competitive bids in real-time. (c) Admin panel's market data entry interface for updating product prices and trends; (d) User registration form for admins to add new users

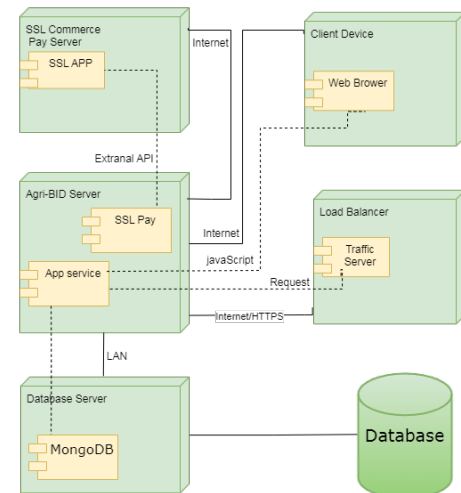


Fig. 6. Deployment Diagram

Fig. 6 depicts the architecture of an agricultural bidding platform with connected components.

This platform is layered in a number of layers. The client device sends its messages on the internet using a web browser to Agri-BID server where the application service and the SSL payment module are hosted. The machine communicates with a back-end oracle database via a secured LAN. In the payment processing, the system is linked to an external SSL commerce pay server using APIs. Javascript creates real-time responsive interaction between the client and the server. This architecture will guarantee modularity, safe processing of payments, and effective access to data.

IV. RESULT AND ANALYSIS

The proposed Smart Bidding Based Agri Trade Platform is an efficient solution to the stated research questions, due to the incorporation of the flexible selling mechanisms, real-time bidding, and advance sell functions (RQ1), as has evidence in other studies [5], [9], and [16]. To provide digital inclusion (RQ2) the system will be mobile responsive, localized content and role-based access [2]. Furthermore, inventory management, agent-based systems, and real-time analytics are associated with making profits and reducing waste (RQ3), which is supported by prototyping the works of [6], [15]. In terms of features, the platform shows much similarity to solution-based literature in terms of authentication, market insight systems, order tracking, and administrative surveillance.

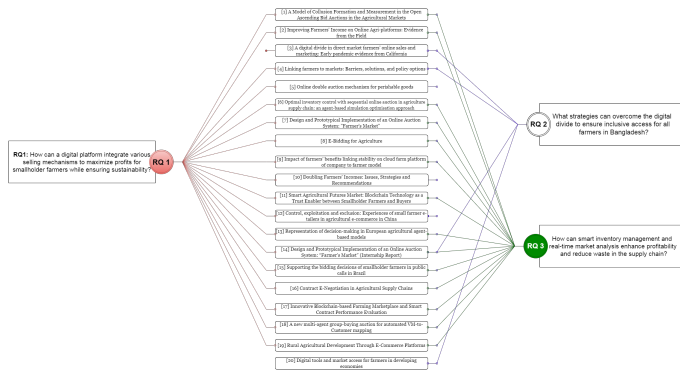


Fig. 7. Research Questions and Paper Mapping

Fig.7 shows an RQ mapping linking research to digital selling, the digital divide, and inventory management in agriculture.

The platform features are Product Listing and Bidding, Action System, Advance Sale Option, Inventory Management, Market Insights, Order Management and Admin Dashboard. The first question RQ1 investigates the ability of other selling processes (e.g. fixed-price, auctions and advance sales) as well as added functionality to maximize the profits of farmers and minimize discard due to better trade planning. RQ2 is devoted to the intersection of the admin dashboard and the integrity of the system, namely, governance, content moderation, and performance monitoring.

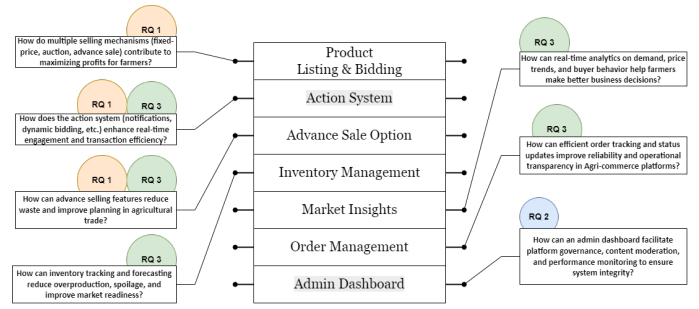


Fig. 8. Research Questions and Feature Mapping

Fig. 8 maps agri-commerce features to three key research questions on trade efficiency and sustainability.

RQ3 investigates how real-time analytics and order tracking as well as dynamic engagement tools could lend the farmers the power to reach informed decision-making, stronger operational visibility, and efficiency in transactions. This congruence attests to the opportunity of coherent online solutions to solve severe dilemmas in agri-commerce environs.

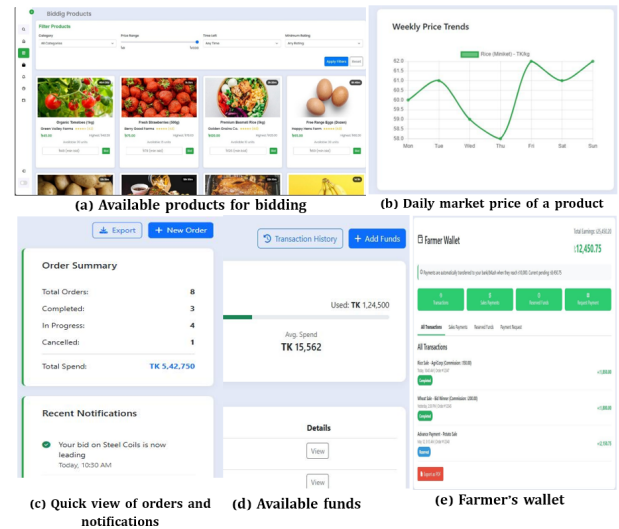


Fig. 9. Output Design Screen

Fig. 9. Feature-Specific Output Interface Screens: (a) Available products for bidding; (b) Daily market price of a specific product; (c) Quick view of orders and notifications; (d) Available funds; (e) Farmer's wallet.

The Smart Bidding Agricultural Platform boosts farmers' economic sustainability by enabling direct sales to buyers, eliminating intermediaries, and improving price discovery, which increases profitability and gives farmers more control over their pricing. Environmentally, it reduces waste and carbon emissions by optimizing inventory management and minimizing transportation. Social sustainability is achieved by empowering farmers with greater bargaining power and access to real-time market insights, allowing them to negotiate better prices and make informed decisions. However, challenges such as limited digital access for farmers and potential collusion

among buyers need to be addressed. Future upgrades will focus on fraud detection, data privacy, and scalability, while pilot testing will provide measurable results like user growth and profit margins.

The system addresses digital inclusion with mobile-friendly training, app usage sessions, and security awareness, empowering farmers and boosting platform adoption. It offers a secure, scalable platform with real-time updates and efficient data handling to streamline transactions and improve market access.

TABLE I
FEATURE MAPPING OF EXISTING RESEARCH

Paper	PLB	AS	ASale	Inv	MI	OM	AD
Das et al. [1]	Yes	Yes	No	No	Yes	No	No
Levi et al. [2]	Yes	Yes	No	No	Yes	No	No
Pesci et al. [3]	No	No	No	No	Yes	No	No
Wanglin et al. [4]	Yes	No	No	No	Yes	No	No
Miyashita et al. [5]	Yes	Yes	No	No	Yes	No	No
Huang et al. [6]	Yes	Yes	Yes	Yes	Yes	Yes	No
Mandhare et al. [7]	Yes	Yes	No	No	No	No	No
Banu et al. [8]	Yes	Yes	No	No	Yes	No	Yes
Fang et al. [9]	Yes	No	No	No	Yes	No	No
Kumarathunga et al. [11]	Yes	Yes	No	No	Yes	No	No
Feng et al. [12]	No	No	No	No	Yes	No	No
Ealias et al. [13]	Yes	Yes	No	No	No	No	No
Alves et al. [14]	Yes	Yes	No	No	Yes	No	No
Bacarin et al. [15]	Yes	Yes	No	No	Yes	No	No
Leduc et al. [16]	Yes	Yes	No	No	Yes	No	No
Varekanand et al. [17]	Yes	No	No	No	Yes	Yes	Yes
Xie et al. [18]	No	No	Yes	Yes	Yes	Yes	No
Our Research	Yes	Yes	Yes	Yes	Yes	Yes	Yes

PLB: Product Listing & Bidding, **AS:** Auction System, **ASale:** Advance Sale, **Inv:** Inventory, **MI:** Market Insights, **OM:** Order Mgmt, **AD:** Admin Dashboard

V. CONCLUSION

Smart Bidding Based Agri Trade Platform is a digital platform meant to assist the agricultural markets with small-holding farmers. It provides a complete digital sales platform with support of fixed price sales, auctions, and pre sales and it enables farmers to have more control over their selling operations and be less dependent on the middle people. The features entailed in the design of the platform include intelligent inventory management, real-time market analysis, and an effective auction system which make market very efficient and less wasteful. It would also guarantee sustainability of the environment maximizing on inventory management thus minimizing over production and food waste. The platform creates a sense of empowerment and strength by increasing the economic independence of the farmers and enabling them to control a greater part of the value of their products. It has a flexible and scalable infrastructure that guarantees its future sustainability, making it a stable solution to unending enhancement in agriculture-trade. The future work will concentrate on streamlining the model, widening its scale, and tackling issues such as the digital divide in achieving the goal of the full potential of the platform.

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